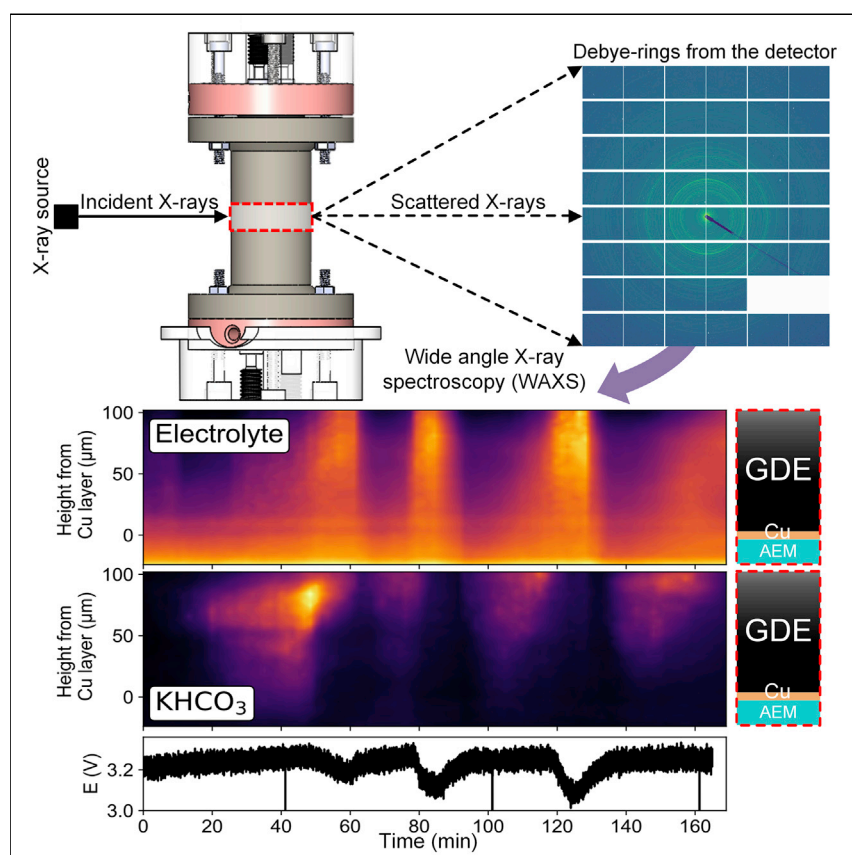


Article

In operando investigations of oscillatory water and carbonate effects in MEA-based CO₂ electrolysis devices



Membrane electrode assembly CO₂ electrolyzers allow commercially relevant current densities. However, there is limited understanding on why or how CO₂ electrolysis shifts toward hydrogen production. Herein, *operando* X-ray scattering reveals a direct observation of salt precipitation and electrolyte flooding in the copper gas diffusion electrode (GDE). The observed oscillations in selectivity and cell potentials are initiated by salts blocking the GDE pores and thus limiting the CO₂ accessibility at the catalyst layer and worsening electrolyte flooding temporarily.

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Highlights

Gradual increase of electrolyte content in the GDE during CO₂ electrolysis

Oscillations are introduced by KHCO₃ salt blocking GDE pores and limiting CO₂ transport

Temporary HER increase correlates with a decrease in cell potential and anode CO₂/O₂ outlet

Salt-induced flooding dissolves precipitates and allows CO₂ recovery at the catalyst

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Article

In operando investigations of oscillatory water and carbonate effects in MEA-based CO₂ electrolysis devices

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SUMMARY

Membrane electrode assembly (MEA) CO₂ electrolyzers are a promising approach toward producing carbon-neutral chemicals; however, they often have stability issues related to flooding of the gas diffusion electrode (GDE). Thus, there is an urgent need to comprehend water management in these devices and to engineer electrodes that allow both stable and efficient electrocatalytic performance. Here, we investigated the possible causes of suppression in the CO₂ reduction reaction (CO₂RR) selectivity on Cu via *in operando* X-ray diffraction (XRD) analysis. The *in operando* XRD allowed us to monitor electrolyte evolution and bicarbonate formation in the GDE, while in-line gas chromatograph and mass spectrometer allowed us to correlate those changes to the cathode and anode product distribution during CO₂ electrolysis (CO₂E). We found direct evidence for salt precipitation in the cathode GDEs, which causes electrolyte build-up and an increase in hydrogen evolution reaction (HER). We also observed that the increase in HER is related to a drop in cell potential, at least partially caused by a shift in ion transport through the membrane from carbonates to more conductive hydroxide ions. Thus, this work suggests that proper ion management is an important key to enhanced durability throughout the device.

INTRODUCTION

Storing renewable energy into chemical bonds/fuels via CO₂ electrolysis (CO₂E) is a promising approach to creating a carbon-neutral cycle while also curbing net CO₂ emissions. At present, copper (Cu)-based membrane electrode assembly (MEA) electrolyzers allow a high rate of CO₂E to $\geq$ C₂₊ products.¹ However, the lack of stability, mainly seen as CO₂ reduction reaction (CO₂RR) being suppressed by the competitive hydrogen evolution reaction (HER), is a major factor delaying the deployment of CO₂E technology on a commercial scale.^{2–6} Therefore, for the future development of these electrolyzers, it is crucial to investigate the exact causes of HER increase over time.

In recent years, many attempts have been made to determine what can cause the observed gradual selectivity change from CO₂RR to HER in a CO₂ electrolyzer.^{2,3,6–8} Overall, it is believed that the issues arise from improper water management and the local reaction environment in the vicinity of the cathode.⁹ For example, sufficient water is necessary to drive the CO₂E, but excess water can flood the gas diffusion electrode (GDE), which restricts CO₂ mass transfer and shifts selectivity to HER.⁵ The reductive potentials needed for CO₂RR lead to electrowetting of the GDE, which

CONTEXT & SCALE

The advancement in sustainable energy technologies is desperately needed to lessen anthropogenic greenhouse gas emissions and mitigate climate change. CO₂ electrolysis is a promising technology to produce value-added chemicals and fuels with renewable electricity, thereby promoting the use of renewable energy into industries. Generally, CO₂ electrolyzers suffer from poor stability owing to flooding of cathode and formation of salt precipitates, causing an increase in competitive hydrogen evolution reaction (HER). However, there is limited understanding of how and why cathode floods and salt deposits occur. We use *in operando* X-ray analysis to identify possible causes of the increase in HER at the cathode interface. We found that it is the salt formation that exacerbates the electrolyte flooding, causing a shift in CO₂ reduction reaction selectivity toward HER. Our X-ray analysis also demonstrated that salt precipitated as bicarbonates despite carbonates being the dominating ionic species transporting through the membrane.

provides a flooding mechanism.^{2,7,10,11} Moreover, the highly alkaline environment created from the cathodic reactions will also equilibrate with CO₂ to form bicarbonate/carbonate (HCO₃⁻/CO₃²⁻) ions that easily permeate through anion exchange membranes (AEMs).^{12–14} Besides lowering the CO₂ utilization, the CO₃²⁻ formation also exacerbates CO₂ mass transport limitation at the cathode.^{12,15} In addition, the hydroxyl groups produced at the cathode (from CO₂RR) can attack hydrophobic supports of the AEM, entailing potential durability issues. All these factors complicate the investigation of catalytic properties as it entangles CO₂ mass transport limitations, ionic conductivity, and water management.

Most of the cathode-flooding mechanisms and concepts are based on fuel cell works^{16–18} with only a few CO₂E studies showing indirect evidence of flooding in a flow cell (i.e., flowing catholyte between GDE and membrane) and zero-gap MEA-based CO₂ electrolyzers.^{2,3,19,20} For instance, Leonard et al.² demonstrated both HER and electrochemical double-layer capacitance (DLC) of the cathode GDE increased with the amount of charge passed through the GDE, where a higher DLC was attributed to a higher rate of cathode flooding. The Berlinguette group measured *operando* water transport aided by numerical simulations in a CO₂ electrolyzer and showed that the major source of water at the cathode varied from being inlet CO₂ feed (when using a 100% humidified CO₂ gas) to the membrane with water diffusing from the anode (when using a dry CO₂ feed).⁶ In another work from the same group, the authors demonstrated that flooding at the cathode can be mitigated by employing thinner membranes (allowing lower net water flux from the anode as compared to thicker membranes) with hydrophobic cathodes at 200 mA·cm⁻².⁵

Although there is a consensus that cathode flooding affects the CO₂RR/H₂ selectivity, an *in operando* visual inspection of a cathode GDE only allows the back surface to be analyzed. To the best of our knowledge, there has yet to be a direct *in operando* approach to show the flooding effect and its relation to product performance. Obtaining *in operando* results is especially important in CO₂E since the local reaction environment varies with time and electrolysis conditions.¹² Furthermore, how a catalyst behaves and whether it changes due to these variations has also yet to be analyzed *in operando*.

Another issue affecting CO₂RR/H₂ selectivity relates to salt depositions in the cathode GDE, which block gas flow to the catalyst.²¹ There is a current debate as to if the salt deposits as alkali metal cation (K⁺/Cs⁺) carbonate^{22–24} or bicarbonate²¹; however, this has only been investigated by computational models²⁴ or postmortem analysis.²⁵ Furthermore, the three-way relationship between flooding, salt deposition, and variations in selectivity has rarely been investigated in depth.

In this work, we couple high flux fourth generation synchrotron X-ray source with superior phase contrast of diffraction phenomena to visualize the interior of an operational GDE to not only understand the relationship between selectivity, flooding, and salt deposition but also to determine how this affects potential (at a constant current) and ion selectivity crossing through the membrane to the anode. Furthermore, this work analyzes structural changes within the Cu catalyst layer, such as how quickly it transitions from a surface oxide to metallic Cu and whether it ripens, gets strained, or migrates within the GDE. It should be noted that the Bazylak group^{19,20} has previously attempted to study both MEA and catholyte-based CO₂ electrolysis devices by X-ray radiography with a focus on gas bubble formations in the MEA and electrolyte layer, respectively. However, they were unable to follow

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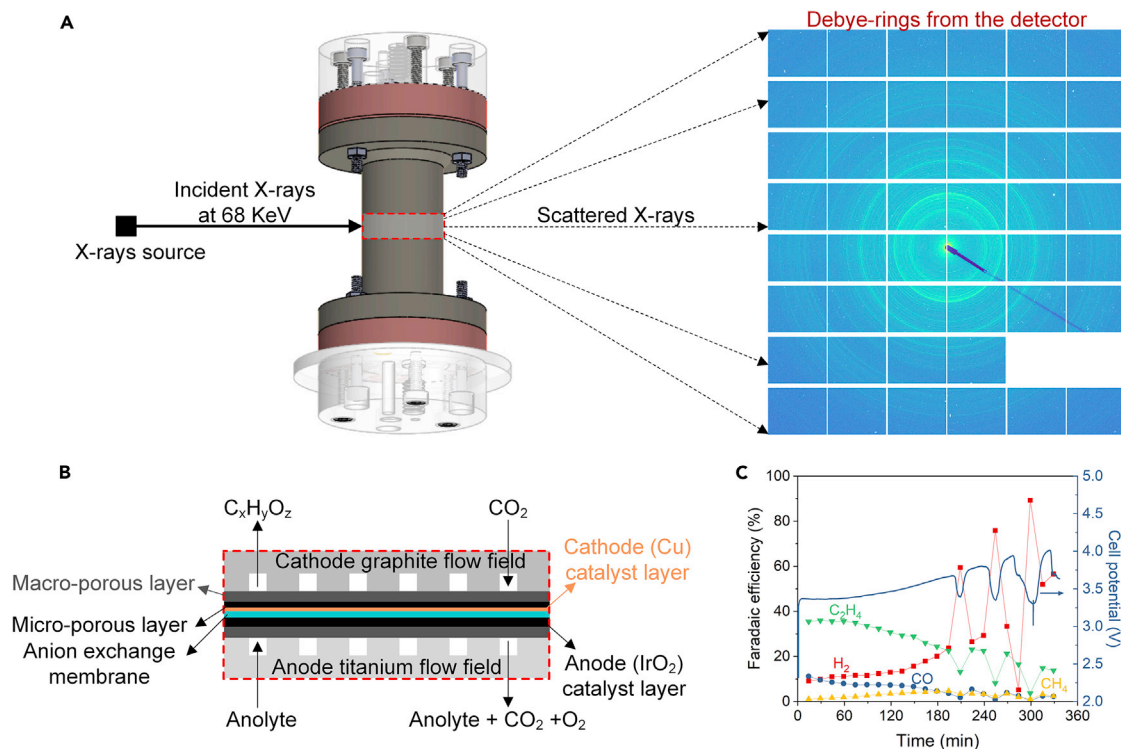


Figure 1. Sketch showing the *in operando* wide-angle X-ray scattering (WAXS) characterization of MEA CO₂ electrolyzers

(A) Scheme illustrating the design of the MEA electrolyzer and the 2D diffraction pattern.

(B) Enlarged view of the highlighted area in (A) where X-rays enter into the electrolyzer. The incident X-ray beam size was $5 \times 20 \mu\text{m}^2$.

(C) Faradaic efficiency (FE) of gaseous products of sputtered Cu GDEs at $200 \text{ mA} \cdot \text{cm}^{-2}$ in a conventional electrolyzer, without X-ray radiation. The experiment was performed with an inlet flow of 30 sccm of dry CO₂ on the cathode and 0.1 M KHCO₃ as an anolyte with a Sustainion (X37-50-RT) AEM.

the evolution of different phases in the catalyst and to concomitantly analyze products, thus limiting the scope of their work.

One additional parameter that this work investigates is the oscillations that are often seen in CO₂ electrolysis devices devolving from CO₂RR to HER selectivity at the cathode. Oscillations typically occur at conditions on the borderline of stability, and thus while not typically consistent, they do provide transient conditions that allow for a much easier understanding of the processes leading to a loss in CO₂RR selectivity.

RESULTS AND DISCUSSION

Reactor performance

Figure 1 shows the reactor and measurement design: the synchrotron X-ray beam ($5 \mu\text{m}_{\text{vertical}} \times 20 \mu\text{m}_{\text{horizontal}}$) probed from the edge of the membrane to deep within the cathode GDE ($\sim 150 \mu\text{m}$ in total) by a continuous series of vertical-line scans allowing for a comprehensive understanding of electrolyte, salt, and Cu as a function of depth within the GDE. To accelerate durability issues related to a change in selectivity from CO₂RR to HER and limited time availability at the synchrotron, we did not use a polytetrafluoroethylene (PTFE) gas diffusion layer²⁶ and low anolyte electrolyte concentrations²⁷ (e.g., 0.01 M) but rather a moderately hydrophobic carbon-based gas diffusion layer and a high-concentration anolyte (0.1 M). Similarly, we used dry CO₂ instead of humidified CO₂.^{6,28,29} Furthermore, we performed a series of *operando* experiments at different current densities (100, 150, 200, and $250 \text{ mA} \cdot \text{cm}^{-2}$) to show that these trends were comprehensive and not a function of a given current density regime.

As a reference, Figure 1C shows the concentrations of gaseous products from one of our preliminary tests conducted using a conventional reactor design (used in our earlier studies)^{13,30} with an active cathode electrode size of 2.89 cm^2 at $200\text{ mA}\cdot\text{cm}^{-2}$ over a $\sim 6\text{ h}$ test period. In this experiment and all other experiments using the X-ray cell, the total faradaic efficiency (FE) for gas products did not sum to 100% due to a significant amount of liquid products.

Liquid products were collected at the end of the experiment, and Figure S1 shows the average FE for all products. The total sum is just below 80% because the total faradaic efficiency is a convolution of the entire experiment and because some of the liquid products diffused to the anode, where they get oxidized. Nevertheless, the gas product distribution provides insights into the reactor's electrochemical performance. The initial performance of the X-ray cell is similar to the conventional cell, both with respect to potential and selectivity. While other works have proposed X-ray irradiation may influence CO_2RR performance,²⁰ we do not see any significant influence on performance related to X-ray irradiation (Figure S2).³¹

For all experiments, including the reference experiment with a conventional cell, the only hydrophobic barrier used was the GDE, and a decrease in overall CO_2RR was expected.²⁶ Figure 1C notably demonstrates a slow HER increase and C_2H_4 and CO decrease during the first 3 h of the experiment. However, beyond 3 h, we found an oscillating trend in FE of gaseous CO_2RR products and HER, correlating well with the cell potential. These oscillations at the increased time were observed in approximately half of all samples tested, both when using a conventional cell and the X-ray cell. With oscillations often related to stability limits, this typically entails an inconsistent behavior: oscillations would occur after some time, but the frequency and intensity (in terms of voltage and productivity changes) would often vary. Figure S3 shows cell-voltage variation over time in multiple CO_2 electrolysis experiments conducted with the X-ray cell at $200\text{ mA}\cdot\text{cm}^{-2}$, confirming a high probability of seeing the oscillating phenomenon.

Our focus here is not to study electrolyzer stability: rather, it is to investigate what causes the increase in HER selectivity and the observed oscillating trends. To do that, we performed a series of *in operando* diffraction measurements using high-energy X-rays. By setting the current density and simultaneously measuring voltage, cathode, and anode selectivity, the state of Cu (oxidation level, micro-structure, particle size, and location), electrolyte content, and presence of crystalline KHCO_3 and K_2CO_3 via a variety of techniques, we gained unprecedented insights into the internal functioning of CO_2 electrolyzers. While we expected flooding to be the leading cause of degradation, we wanted to ensure that the loss in CO_2RR selectivity did not arise from a change in the catalyst itself. To confirm this, we provide a detailed analysis in Note S1 (including Figures S4–S11) on how the Cu catalyst did not change (only native Cu_2O reduced to metallic Cu) and how there was no IrO_2 crossover from the anode side to the cathode during CO_2 electrolysis.

Flooding of cathode GDE

The electrolyte distribution in the different parts of the cell, mainly GDE, is followed by deconvoluting the liquid phase diffraction pattern from the diffractograms.^{32–34} We took advantage of the fact that the other solid amorphous phases did not change during the experiment, and any change in the q -range 2.45 to $2.46\text{ \AA}^{-1}$ of the WAXS pattern therefore must have been caused by scattering of electrolyte. The plot in Figure 2B was made by integrating the scattering intensity in the given q -range.

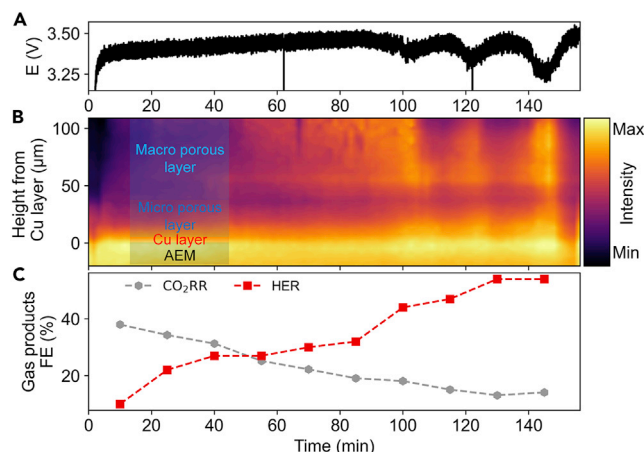


Figure 2. Evolution of electrolyte content and CO_2RR selectivity in the GDE over electrolysis time at $200 \text{ mA} \cdot \text{cm}^{-2}$

Change in (A) cell potential (E (V)), (B) electrolyte content, and (C) product selectivity of gaseous CO_2RR products and HER over electrolysis time. The experiment was performed at $200 \text{ mA} \cdot \text{cm}^{-2}$. A detailed version of the electrolyte plot and gaseous CO_2RR product selectivity is presented in Figures S14 and S15 of the supplemental information. In (B), the change in color intensity from black to yellow indicates an increase in X-ray signals corresponding to the electrolyte content.

The integrated values from all obtained patterns were then plotted as a function of corresponding height and time. Figure 2B gives a qualitative visualization of the electrolyte content at different heights as the experiments ran. The color of the heatmap corresponds to the intensity of the integrated signal, and thereby the electrolyte content, ranging from an arbitrary minimum (black) to arbitrary maximum (yellow). Investigations of individual patterns confirm that the changes observed in the full q-range background match well with water/electrolyte. For example, we show some diffraction patterns (Figure S12) and change in the background (Figure S13) at a single region in the GDE during different electrolysis time.

As seen in Figure 2, the overall electrolyte content in gas diffusion layer (GDL) increased slowly until ~ 100 min where it suddenly decreased. Given that the selectivity change correlates with the electrolyte content in the GDL, this slow increase of electrolyte content in the GDL likely explains why we see a slow degradation in CO_2RR performance over time. For example, with GDL pores slowly filled with electrolyte, the CO_2 diffusion distance toward the catalyst layer increases, reducing CO_2 availability and causing a gradual shift from CO_2RR to HER. Interestingly, the electrolyte content is observed further toward the flow field than we expected. Recent studies reported that loss of GDE hydrophobicity during electrolysis (in addition to the formation of liquid products affecting the physical properties of the GDE) can cause the electrolyte to penetrate deeper into the GDE.^{2,3,7} The electrolyte content seems to be highest in the outer layers of the GDL ($>50 \mu\text{m}$), whereas, there seems to be a relatively unchanged region closer to the catalyst layer (Figure 2B). A similar trend is observed for all GDEs, and we expect this to be caused by the differences in PTFE content (25% versus 5%) and pore size ($<1 \mu\text{m}$ versus $25\text{--}200 \mu\text{m}$) of the micro- and macro-porous layers of the Sigracet 39 BB GDL, resulting in different water management properties.^{35,36} The degree of penetration of the microporous layer into the macroporous layer as well as GDL compression are unknown; thus, an exact determination of the microporous/macroporous transition cannot be determined. Nevertheless, this order of magnitude thickness does align with a relatively dry area between the catalyst and outer GDL layers.

Initially, the slow increase in electrolyte content and the cell potential follow until around 90 min. Both correlate with a steady increase in HER, indicating that slow flooding is causing the observed performance degradation. From around 90 to 100 min, the HER seems to increase faster, but at this point, the potential starts to decrease while the electrolyte content continues to increase. The initial correlation breaks at this point, suggesting that a different mechanism is accelerating the performance degradation. Once the potential reaches a local minimum (~ 100 min), the potential starts to increase again. Shortly after (~ 105 min), the electrolyte content drastically decreases. The oscillation repeats in roughly 20 min cycles with a 3–5 min delay between the minimal potential and maximum electrolyte content.

Such oscillations, both in terms of cell potential and electrolyte content in GDL, could be caused by changes in the anode potential, e.g., from anode degradation or bubble formation, or could be due to variations in gas pressure and temperature in the electrolyzer. There are, however, no indications that any of these are the cause. We measured the anode potential by putting a reference electrode in the inlet flow channel of the anolyte. The anode potential remained stable throughout the CO_2 electrolysis time, indicating that any change in the cell potential is either arising from the cathode potential or ohmic losses in the membrane (Figure S16). Small periodic variations in the cathode inlet pressure (Figure S17) were observed (with different periodicity) due to a slight overpressure in the gas chromatograph (GC). To confirm that the GC caused overpressure, we did a control experiment where the cathode outlet was not connected to the GC and observed no change in the cathode inlet pressure (Figure S18). Also, no variation in cathode flow-field temperature was observed during the electrolysis (Figure S17). Compared to the reference experiment with a conventional cell (Figure 1C), the CO_2RR stability is significantly lowered in the X-ray cell (Figure 2B). This was generally observed when using the X-ray cell and is expected to be caused by the smaller electrode area, different flow channel geometry, and a non-identical GDE compression. Since accelerated degradation was desired for *operando* experiments, no attempt to improve the stability was made. Despite the differences, the overall trend and oscillating behavior were observed with all cell types, but their oscillations and CO_2RR stability varied significantly, pointing to the chaotic nature of the processes driving these fluctuations.

Figure 2C shows the faradaic efficiency of the cathodic gas products (Figure S15 shows the exact CO_2RR product breakdown from Figure 2). The FE of H_2 gradually increases during the first 80 min before further rapid increase at 100 min. For the next 15 min, the rate of H_2 selectivity increase slows down slightly before the start of the second cycle of acceleration and then the deceleration of the H_2 selectivity increase. The slow sampling rate of the gas chromatograph dilutes the trends, but there is a notable correlation between the electrolyte content in the GDE and the level at which HER increases.

The shift to a lower potential during the spike of H_2 selectivity provides an intriguing result, contrary to what is seen in the first 80 min (Figure 2C). Here, there is a gradual increase in operating potential while HER slowly increases. Under the constant current, and at the studied conditions, CO_2RR dominates the selectivity. The reason for CO_2RR selectivity cannot be due to mass transfer limitations of water for HER (since CO_2RR also needs water) but must be due to catalysis. Thus, with CO_2RR being a more effective catalysis, the HER would need a higher potential to maintain the same current as CO_2RR . While this argument agrees with the first 80 min of the experiment, during the fluctuations, an additional factor must be in play causing a higher HER and lower voltage. For example, an ohmic resistance decrease in the

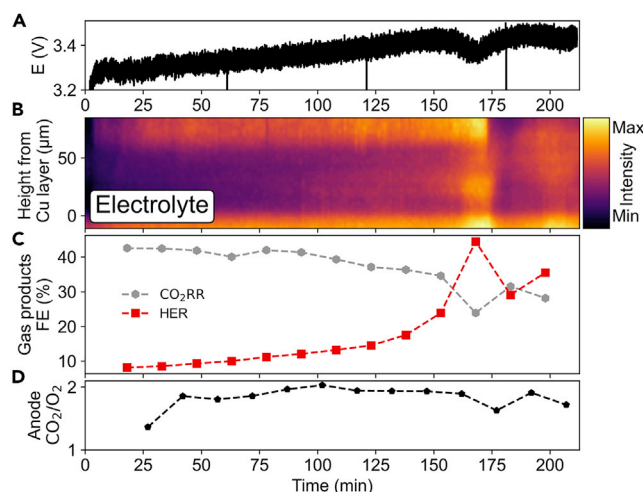


Figure 3. Evolution of electrolyte content, CO₂RR selectivity, and CO₂ to O₂ ratio in the GDE over electrolysis time at 150 mA · cm⁻²

(A–C) Change in (A) cell potential (E (V)), (B) electrolyte content, and (C) product selectivity of gaseous CO₂RR products and HER.

(D) CO₂/O₂ ratio at the anode as a function of the electrolysis time.

The experiment was performed at 150 mA · cm⁻². In (B), the change in color intensity from purple to yellow indicates an increase in X-ray signals corresponding to the electrolyte content.

membrane (or other places within the device) or fundamental variation in catalytic activity could be the cause of lower cell potential and increased HER. The conductivity of the ion exchange membrane is well known to be a function of the ion being transferred. For instance, the conductivity of a Sustainion membrane X-24 membrane in KHCO₃ is 24 mS · cm⁻¹, whereas the same membrane in KOH shows a conductivity of 66 mS · cm⁻¹.³⁷ While this work uses a different version of Sustainion, we expect the trends to still hold. It is well established that analyzing the anodic CO₂ to O₂ ratio in CO₂ electrolysis devices works as a proxy to determine what species (i.e., HCO₃⁻, CO₃²⁻, or OH⁻) are transferring through the membrane.^{12,13,38} A CO₂ to O₂ ratio of 4 means HCO₃⁻ transfer, a ratio of 2 means CO₃²⁻, and a ratio of 0 means OH⁻ transfer.

To understand if the ion transport across the membrane could be contributing to potential oscillations, we investigated the anodic CO₂ to O₂ ratio. All experiments indicate a decreased potential correlates with a lower CO₂ to O₂ ratio, but the trend is seen more clearly in some experiments. Figure 3 shows an experiment performed at 150 mA · cm⁻² (Figure S19 shows a detailed version of Figure 3B). The CO₂ to O₂ ratio for the other experiments can be found in the supplemental information. Again, we see the link between electrolyte content, cell potential, and faradaic efficiency of H₂, albeit much more clearly. Figure S20 shows detailed cathodic gas product distribution from the GC, whereas Figure S21 shows cathodic and anodic gas products from a mass spectrometer operated concurrently with the GC. In Figure 3D, we can now see that the CO₂ to O₂ ratio is initially near 2, entailing carbonate transfer, which is expected of a device with a well-functioning cathode.^{12,13} However, when we start seeing a change in potential, we see a drop in the CO₂ to O₂ ratio, entailing a partial switch from a CO₃²⁻ transfer to an OH⁻ transfer. The OH⁻s conduct more efficiently through the membrane, which partially explains the drop in potential. Furthermore, at the point when the membrane again becomes more CO₃²⁻-rich (i.e., CO₂:O₂ ≈ 2) the potential also increases again. It should be noted that the CO₂:O₂ ratio is a

function of both analyte equilibration time and GC measurement time, thus it is hard to quantify the lag between the $\text{CO}_2\text{:O}_2$ ratio and potential (also, there was a lag of 9 min between the cathode and anode gas injection to the GC).

Another point to note is that during CO_2 electrolysis, the surface is covered by a certain percentage of CO, and thus this is known to decrease HER catalytic activity substantially.³⁹ In the case where the catalyst is CO_2 deficient, this will entail enhanced activity and should result in a lower device potential.³⁹ Given that a CO_2 deficiency will result in both an increase in membrane conductivity and a catalytic improvement for HER, these two parameters are intrinsically coupled, thus concluding the relative impact of each is beyond the scope of this work (especially since this involves a chaotic oscillatory system being analyzed).

If the fundamental issue is mainly flooding of the cathode, this would dilute the concentration of cathodically produced OH^- s. The dilution of OH^- s would lower the pH, entailing a shift toward HCO_3^- instead of CO_3^{2-} and an increasing $\text{CO}_2\text{:O}_2$ ratio approaching 4.³⁸ However, Figure 3D shows just the opposite trend: this decrease can only occur by a lack of CO_2 , entailing OH^- transfer through the membrane since this case would have a $\text{CO}_2\text{:O}_2$ ratio of 0. Another exciting observation, though less clearly visible, is that the potential and selectivity shifts seem to be slightly preceding the flooding. This effect was seen in all synchrotron experiments where fluctuations were clearly visible (see Figures S22–S25 for additional experiments). As it is well known that salt precipitation occurs during CO_2E and degrades performance,^{21,23} our following experiments focused on monitoring salt formation and whether we could conclude that this was the instigator of the oscillations.

Salt precipitation in the cathode GDE

Salt depositions are in crystalline form and thus can easily be isolated via their diffraction pattern (see supplemental information for details). By operating another experiment, this time at $100 \text{ mA} \cdot \text{cm}^{-2}$, we found distinct oscillations while monitoring electrolyte and carbonate content, as shown in Figure 4. Unfortunately, we do not have cathodic GC or MS data for this synchrotron experiment (due to a technical issue); however, the anode CO_2 to O_2 ratio (Figure S26) shows trends consistent with previous experiments.

One of the most interesting characteristics of Figure 4 is that salt precipitation occurs preceding the substantial increase in electrolyte content. These results are in contrast to previous reports that assumed salt precipitation is a result of massive electrolyte flooding.^{7,8} It is even more interesting that KHCO_3 was found (as validated by all its XRD peaks), but we found no XRD peaks attributed to K_2CO_3 . Figure S27 shows an example of a wide-angle X-ray scattering (WAXS) pattern obtained 91 μm above the Cu layer at 103 min into the experiment shown in Figure 2 and reference patterns for KHCO_3 and K_2CO_3 . Previous reports^{22–24,40} have indicated seeing K_2CO_3 at the cathode, and given that it is known that CO_3^{2-} is primarily transferring through the membrane (see Figures 3 or S26), it would be expected that any liquid electrolyte at the cathode would be in the form of K_2CO_3 . While it is not clear why KHCO_3 deposits to begin with instead of K_2CO_3 , one hypothesis is that there become localized areas of more neutral pHs due to higher CO_2 concentrations near the flow field, thus allowing the formation of KHCO_3 , and since the solubility of KHCO_3 is less than half that of K_2CO_3 ,⁴¹ this leads to the salt deposition.

As we start (before electrolysis) with no potassium ions (K^+) at the cathode and the solubility limit of both KHCO_3 (3.62 M at 25°C)⁴¹ and K_2CO_3 (8.03 M at 25°C)⁴¹ is

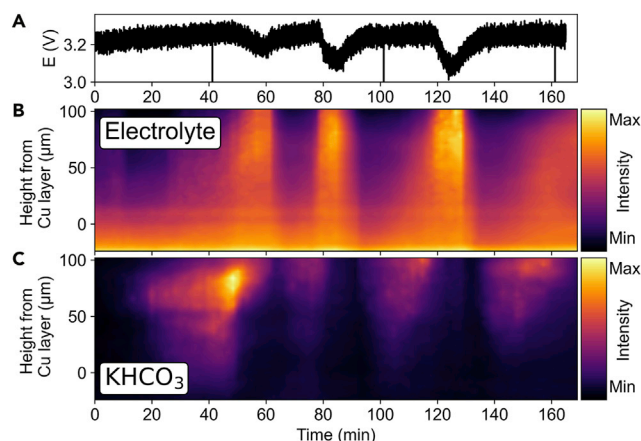


Figure 4. Evolution of electrolyte content and salt precipitates in the GDE over electrolysis time at $100 \text{ mA} \cdot \text{cm}^{-2}$

Changes in (A) cell potential ($E(V)$), (B) electrolyte content, and (C) salt precipitation in the cathode GDE as a function of electrolysis time. The KHCO_3 salts were found by integrating two KHCO_3 peaks at 2.19 and $2.20 \text{ \AA}^{-1}$ and subtracting the background. The plot at the top shows the simultaneous change in full cell potential. This experiment was performed at $100 \text{ mA} \cdot \text{cm}^{-2}$. In (B) and (C), the change in color intensity from purple to yellow indicates an increase in X-ray signals corresponding to the electrolyte content and KHCO_3 salt precipitation respectively. Also, the color intensities shown in (B) and (C) are not directly comparable and must be treated separately.

significantly higher than the 0.1 M KHCO_3 used as anolyte, it is evident that a fair amount of K^+ are transported from the anode to the cathode by electrostatic force. This force will be counterbalanced by diffusion when the concentration at the cathode exceeds the anolyte concentration. The electrostatic force and diffusion will be outbalanced at a particular K^+ concentration in the cathode, and then the net flux of K^+ across the membrane is expected to be zero. This, of course, is under the assumption that K^+ are not diffusing away from the membrane-electrode interface and into the GDE and flowfield, but this is exactly what we observe. The GDE can therefore be regarded as a K^+ reservoir being filled up during the electrolysis reaction. This raises the question about whether the observed oscillations in potential and selectivity could be caused by changes in K^+ concentration either at the cathode or the anode. Given the almost-constant electric field across the membrane (as seen from the almost-constant cell potential in all experiments), it seems unlikely that the electrostatic driving force for K^+ transfer toward the cathode should change significantly during CO_2E . The osmotic driving force is of course changing as the concentration of K^+ at the cathode side increases, but this is only until the concentration reaches the solubility limit. Hereafter, a further K^+ transfer would fill up the GDE, as observed. We, therefore, do not expect any oscillating behavior of the K^+ concentration in the vicinity of the cathode catalyst during the electrolysis. It is, though, relevant to ask if the concentration at the anode is lowered so much that the K^+ flux is affected. Assuming that all of the cathode GDE volume ($\sim 0.015 \text{ mL}$) is filled with KHCO_3 salt, 5.5% of the K^+ in the $60 \text{ mL } 0.1 \text{ M}$ anolyte reservoir would be consumed (see [Note S2](#) for calculation). The concentration of the remaining anolyte would then be 0.094 M .

To exclude that such changes in anolyte concentration can cause oscillations, we performed a control experiment with 0.01 M KHCO_3 as the anolyte ([Figure S28](#)). The experiment shows similar cell potential and selectivity behavior during the first 3 h, as seen in [Figure 1C](#). Slightly higher initial CO selectivity in 0.01 M KHCO_3

(Figure S28) as compared to 0.1 M KHCO_3 (Figure 1C) is possibly from lower initial cation concentration at the cathode with 0.01 M KHCO_3 anolyte.⁴² Further analysis is outside the scope of this work. As expected, oscillations occurred, though slightly later than what was observed in 0.1 M KHCO_3 experiments, indicating any change in K concentration (irrespective of the initial concentration) in the anolyte would not cause the observed oscillations.

While finding an apparent reason for the formation of KHCO_3 is challenging, there are highly plausible reasons for its disappearance once the electrolyte reaches the salt crystals. The most obvious is that the aqueous solution simply dissolves it, but a more subtle point is that this electrolyte is expected to be highly alkaline and lacking carbonates (as denoted by OH^- rather than CO_3^{2-} crossing over the membrane). This alkaline environment (i.e., KOH based) would entail that the KHCO_3 rapidly converts to the much more soluble K_2CO_3 , which would help in solubilizing the KHCO_3 crystals and preventing the dissolved salt from precipitating even when the electrolyte level (in the GDE) is once again lowered.

To further confirm that salt precipitation is triggering the observed oscillations in CO_2RR , we did multiple CO reduction (COR) experiments. As expected, we do not see any oscillating phenomenon in cell potential and COR products/HER selectivity (Figure S29). In addition, no signs of salt precipitation were found in the cathode flow field post COR.

Comprehensive analysis of oscillations

Based on the preceding analysis, we can link many of these effects to provide a comprehensive mechanism for the oscillatory nature seen during CO_2 electrolysis. Figure 5 summarizes the mechanism that is plausible based on this analysis. It is important to note that some of these processes often take place simultaneously, and the order in which they are taking place may be slightly different than proposed. Nevertheless, this does give an overview of the major processes taking place.

Initially, we observe a small amount of electrolyte deep into the GDL, and we see the formation of KHCO_3 salt crystals. These salts cause a blockage in the GDE, preventing sufficient CO_2 access. We base the hypothesis that the salt crystals are blocking the CO_2 access on the fact that we see both a decrease in CO_2 reduction selectivity in favor of HER and a decreased carbonate crossing over the membrane. Thus, from a pure mass transfer standpoint, there is a loss of CO_2 reaching the catalyst/membrane interface. This CO_2 deficiency at the cathode leads to anions transporting through the membrane shift to OH^- , causing a drop in membrane ohmic losses and thereby decreasing the overall cell potential. Additionally, the lack of CO_2 and its intermediate such as CO bound to the catalyst also increases catalytic activity for HER and decreases the cell potential. Beyond this point, the electrolyte then floods the cathode. Given that the incoming cathode gas is dry CO_2 , this electrolyte must be coming from the anode, but the reason for this increased electrolyte penetration is still unclear. One factor that could lead to increased electrolyte penetration is that membranes in the OH^- form are more hydrated than in the CO_3^{2-} ,⁴³ and this increased swelling could potentially lead to an increased water transfer from the anode to the cathode. This hypothesis is supported by clear observations of local electrolyte content increase inside the membrane (electrolyte heatmaps in Figures 2, 3, and 4). However, more experiments would be needed to verify this effect. Another factor is that osmotic drag naturally pulls water from the anode to the cathode to hydrate the salts. Given the system's complexity, additional factors could not be accounted for; thus, all that can be determined for certain is that the electrolyte returns

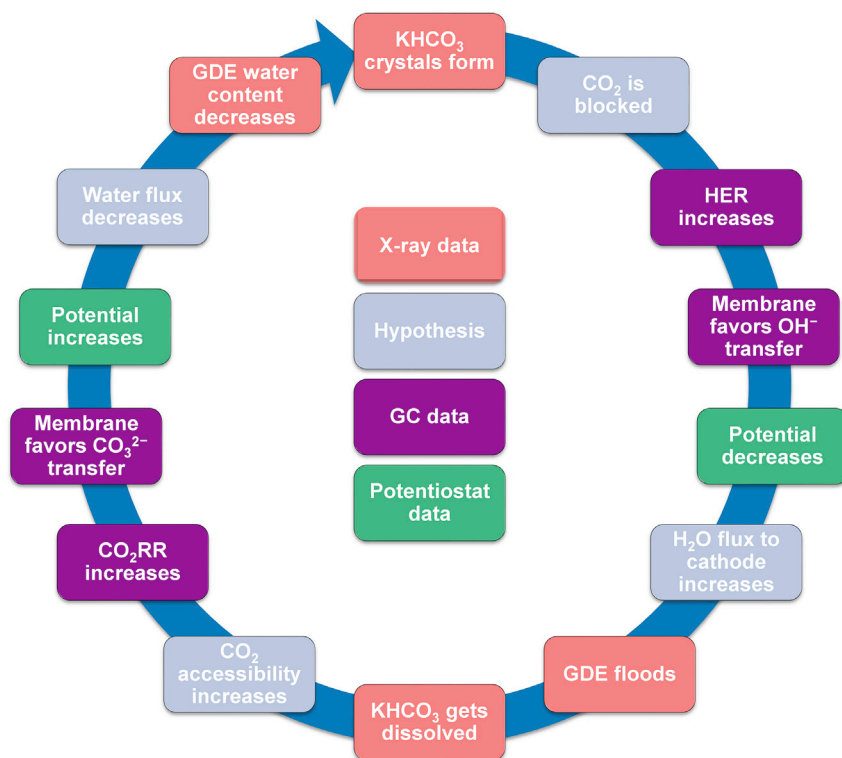


Figure 5. The proposed mechanism illustrates the oscillating phenomenon of salt precipitation and dissolution causing a change in the electrocatalytic performance of the CO₂ electrolyzer

It should be noted that the illustration only describes the oscillations. Underlying performance degradation due to slow flooding from other causes and potassium build-up at the cathode have not been included. The total observed phenomena would therefore be more like a degrading spiral.

to the cathode. The increased electrolyte content at the cathode dissolves the salt and removes the CO₂ blockage; this not only allows for a recovery in CO₂RR but also shifts the ion transport in the membrane back to carbonates, causing the cell potential to recover (or oscillate). It should be noted at these current densities, CO₂ blockage should be the dominating factor as mass transfer effects should be marginal even in partially flooded GDEs.⁴⁴ At the same time, the water flux to the cathode is decreased, causing a dry period in the GDE. Thereby salt can again precipitate, and the cycle shown in Figure 5 starts over.

Conclusion

While the oscillatory effects seen in this work are clearly not desired from an applied standpoint, these oscillations enabled us to gain deep fundamental scientific insight into the processes that lead to loss of selectivity to CO₂ products by analyzing these oscillations via monitoring a wide variety of parameters such as potential, cathode and anode products, Cu oxidation state, as well as location and intensity of Cu, electrolyte content, and salt deposition in the GDE. Although our results showed that Cu is relatively stable (at least at the micro-scale), there is a highly interconnected relationship between electrolyte penetration into the GDE, salt deposition, membrane ion conduction (CO₃²⁻ versus OH⁻), and overall cell potential at a given current density. Though this work opens new questions, it does clarify that mitigating salt deposition throughout the gas diffusion layer is essential to obtaining long-term stability.

Thus, further works on durability should focus on enhanced reactor designs, salt management processes, and more resistant gas diffusion layers to help resolve these issues.

Based on all of our findings, understanding and tailoring transport mechanisms in the AEMs is essential to obtain a stable CO₂E operation. Therefore, future studies should focus on improving the AEM performance, especially ion conductivity, water uptake, water transport, etc. In addition, the transport and diffusion of water, anion, and cation transport through the membrane are apparent areas for further research.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Brian Seger (brse@fysik.dtu.dk).

Materials availability

This study did not generate new unique reagents.

Data and code availability

Raw X-ray data generated at the European Synchrotron Radiation Facility (ESRF) large-scale facility are available at <https://doi.esrf.fr/10.15151/ESRF-ES-439498753> from 2024. Alternatively, these data can be available from the lead contact upon reasonable request.

Electrode preparation and zero-gap MEA setup

The cathode GDEs for all the experiments were prepared by sputtering a 150 nm layer of 6N Cu onto a Sigracet 39BB carbon paper (Fuel Cell Store) in a vacuum environment (10^{-5} – 10^{-6} Torr) at a deposition rate of $\sim 1 \text{ Å s}^{-1}$ under 5 sccm Ar, total pressure 3 mTorr, and at room temperature. The anode GDE was a commercial IrO₂-coated GDE (Dioxide Materials).

All electrochemical CO₂E tests were performed in a 0.64 cm² MEA-based CO₂ electrolyzer. Figure 1 shows a schematic description of the cell (for more details, refer to our X-ray cell article³¹). The MEA is assembled by placing the anion exchange membrane (AEM, Sustainion X37-50, Dioxide Materials) between the cathode and anode. The as-received AEMs were presoaked and activated in 1 M KOH for 24 h and then washed with deionized water (18.2 MΩ). After activation, the AEMs were stored in deionized water before use. The cell consists of a titanium serpentine flow field on the anode side and a graphite serpentine flow field on the cathode side. The two flow fields are inserted in a PEEK housing enabling X-ray diffraction experiments with the incident beam being in-plane with the MEA.

Electrochemical measurements

A dry CO₂ gas (for non-X-ray irradiated samples at DTU (Figure 1C) Linde, 5N, for X-ray irradiated samples at ESRF Air Products, 5N) was supplied to the cathode flow field with a flow rate of 30 sccm using mass flow controller (Vöegtlin red-y smart series), while, the anode was fed with 0.1 M KHCO₃ (Sigma-Aldrich, 99.99% metal basis). Cathode inlet pressure was measured using a pressure transducer (Brooks Instrument, SolidSense II Series). A constant volume of 60 mL anolyte was recirculated through the anode with a digitally controlled diaphragm pump (KNF NF1.5TTDCB-4) set to 25% corresponding to approximately 7.5 mL min⁻¹. The anolyte reservoir was purged with Ar (at 30 sccm) to carry the gases formed at the anode for gas

chromatography analysis. All electrochemical measurements were performed under galvanostatic mode with a potentiostat (Biologic SP-300 and SP-240) and the cell voltages are reported without any iR correction. A leakless Ag|AgCl reference electrode was introduced before the anolyte inlet at the anode side to measure the anodic potential. A schematic of the setup is shown in [Figure S30](#).

Product analysis

Gas product analysis

All cathodic gas products such as H_2 , CO , CH_4 , C_2H_4 , and C_3H_6 products were analyzed using a gas chromatograph (GC, Agilent 6890A) equipped with a thermal conductivity detector, flame ionization detector, and capillary columns (45 m $\times$ 0.530 mm 40.0 Micron HP-PLOT/Q and 30 m $\times$ 0.530 mm 15.0-micron HP-Molesieve). Argon (Linde, 5N) was used as a carrier gas in the GC. The gas chromatograph also analyzed gases such as CO_2 and O_2 at the anode side. The GC was calibrated by injecting a mixture of calibration gas (including N_2 , CO , CH_4 , H_2 , CO_2 , C_2H_4 , and C_3H_6) for analyzing cathodic gas products and a separate calibration gas mixture (including N_2 , O_2 , and CO_2) for analyzing anodic gas products.

The faradaic efficiency (FE) of the gas products was estimated using the following equation:

$$FE = \frac{z_i \cdot F \cdot x_i \cdot \dot{n}}{I} \quad (\text{Equation 1})$$

where z_i is the number of electrons transferred per mole of gaseous product i involved in the CO_2RR , F is Faraday's constant ($96,485 \text{ C mol}^{-1}$), x_i is the mole fraction of gas i in the outlet cathodic gas mixture analyzed using GC, $\dot{n}$ is the molar flowrate in mol s^{-1} , and I is the total current in A. The molar flowrate can be calculated from the volumetric flowrate (Q) by the ideal gas equation ($\dot{n} = P_{atm}Q/(RT)$) where P_{atm} is the atmospheric pressure in Pa, R is the ideal gas constant ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$), and T is the temperature in K. The volumetric gas flow rates (Q) at the outlet of both the cathode and anode were measured by introducing a known concentration of N_2 (Linde, 5N) via a mass flow controller placed right before the inlet of the gas chromatograph.

Liquid product analysis

The outlet cathode gas stream was passed through a gas wash to extract the liquid CO_2RR products. The liquid CO_2RR products both at the cathode (extracted from the gas wash) and at the anode (diffused through the anion exchange membrane) were quantified using a high-performance liquid chromatograph (HPLC, Agilent 1260 Infinity II) with 0.1 M $KHCO_3$ as the reference solution and 0.05 M H_2SO_4 as the internal solvent. For all the measurements, 2 mL of collected samples (anolyte/gas wash at the cathode) were put into vials and used for further quantification. HPLC consists of an autosampler, a quaternary pump (for mobile phase transport and the mixture of different eluents), a column (Aminex HPX-87H packed column with a cross-linked resin in hydrogen form), and two detectors (RID: refractive index detector and DAD: diode array detector). The HPLC was calibrated using known concentrations of different liquid products (e.g., ethanol, formate, acetate, n-propanol, etc.). The developed HPLC method takes 1 h to analyze each sample.

Next, we measured the concentrations (c_i) and the faradaic efficiency (FE) of the liquid products using [Equations 2](#) and [3](#) respectively.

$$c_i = \frac{A_i}{C_{f,i}} \quad (\text{Equation 2})$$

$$FE = \frac{z_i \cdot c_i \cdot V_r \cdot F}{I \cdot t} \quad (\text{Equation 3})$$

where A_i is the peak area of liquid products from the chromatographs obtained in the HPLC, $C_{f,i}$ is the calibration factor for each liquid product i , V_r is the reservoir volume (which is 60 and 30 mL and for the cathodic and anodic liquid products), t is the total electrolysis time in seconds.

In-operando X-ray measurements

Synchrotron WAXS measurements were performed at the ID31 beamline of the European Synchrotron Radiation Facility (ESRF) in Grenoble, France. The high-energy X-ray beam (68 keV) was focused ($5 \mu\text{m}_{\text{vertical}} \times 20 \mu\text{m}_{\text{horizontal}}$) on the MEA in grazing incidence mode (beam parallel to the MEA layers), and the scattered signal was collected using a Dectris Pilatus CdTe 2M detector positioned 710 mm behind the sample. The energy, detector distance, and tilts were calibrated using a standard CeO_2 powder and the 2D diffraction patterns were reduced to the presented 1D curves using the pyFAI software package.⁴⁵ To avoid the high, spotty, and randomly distributed signal arising from the gasket in the cell, iterative filtering has been implemented in the azimuthal integration (referred to as “sigma_clip” integration in the azimuthalIntegrator Module in the pyFAI package).

Rietveld refinements of the WAXS patterns were performed to extract the crystallite sizes, lattice parameters, and microstrains using the Fm3m structure of Cu metal and the GSAS-II software.⁴⁶ The instrumental parameters were determined by the refinement of a CeO_2 standard sample. As repetitive vertical scans across the cathode GDE and the membrane were performed during the experiment, the background subtraction consisted of removing the GDE signal from the catalyst signal.

Electrode characterization

The surface composition of the used cathode GDEs was determined using a Kratos Axis Ultra DLD X-ray photoelectron spectrometer (XPS). All the measured spectra were acquired using a monochromatized Al K_{α} radiation at 15 kV and 15 mA. The binding energy of the acquired spectra was calibrated using C 1s binding energy to 284.8 eV. Ion-scattering spectroscopy profiles of the cathode GDEs were also done to identify any contamination after CO_2 electrolysis.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.joule.2023.01.013>.

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AUTHOR CONTRIBUTIONS

A.B.M. and S.G. carried out all electrochemical measurements, conducted data analysis, and wrote the manuscript with input from all co-authors. M.M. and J.D. assisted in performing *in operando* experiments (including data analysis) at ESRF, and

M.M. wrote the part on how *operando* measurements were performed. C.A.G.R. helped with electrochemical measurements performed at ESRF and liquid product analysis. R.I. helped with performing data analysis for the X-ray data. I.C. and B.S. conceived the project and oversaw its development.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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